Student Information

Information	
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Section	BSCS 4B
Assigned Dataset	A Large Scale Fish Dataset
Kaggle Dataset URL	https://www.kaggle.com/datasets/crowww/a-large-scale fish-dataset?
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Part 1 — Dataset Exploration

1.1 Dataset Description

Brief Description

Describe your assigned dataset in your own words.

The dataset is a large-scale fish image dataset containing 9 fish classes. The notebook loads 9,000 images, resizes them to 64 × 64 pixels, keeps three color channels, and normalizes pixel values to the 0–1 range before training a neural network.

1.2 Dataset Characteristics

Complete the following table based on your actual dataset.

Dataset Characteristic	Result
Number of Classes	9
Total Number of Images	9,000

Image Format	PNG/JPG/JPEG accepted by the loader; exact dataset mix not reported
Image Dimensions	64 × 64 × 3 after preprocessing
Color Channels	3 (RGB)
Smallest Class	1,000 images per class (balanced dataset)
Largest Class	1,000 images per class (balanced dataset)

1.3 Classes

List all classes found in your dataset.

#	Class Name	Number of Images
1	Black Sea Sprat	1,000
2	Gilt-Head Bream	1,000
3	Hourse Mackerel	1,000
4	Red Mullet	1,000
5	Red Sea Bream	1,000
6	Sea Bass	1,000
7	Shrimp	1,000
8	Striped Red Mullet	1,000
9	Trout	1,000

1.4 Initial Observation

Based on your initial exploration, what did you observe about the dataset?

The dataset contains nine classes and 9,000 images, with an apparently balanced *class distribution. The preprocessing converts every image to the same 64 × 64 × 3 shape and normalizes pixel values. Because the classes include visually similar fish such as Red Mullet and Striped Red Mullet, a simple fully connected network may have difficulty distinguishing some categories.*

Part 2 — Train, Validation, and Test Sets

2.1 Dataset Splitting

The notebook uses a stratified 70 / 15 / 15 split. First, 30% of the data is held out; that temporary set is then split equally into validation and testing sets. Stratification preserves the class distribution across the splits.

Dataset Split

Dataset	Number of Images	Percentage
Training	6,300	70%
Validation	1,350	15%
Testing	1,350	15%
Total	9,000	100%

2.2 Purpose of Each Dataset

Explain the purpose of each set.

Training Set: Used to learn the neural network's weights from labeled fish images.

Validation Set: Used during training to monitor performance on data not used for updating the weights.

Testing Set: Used only after training to measure the model's final performance on unseen data.

2.3 Observation

Why is it important to keep the test dataset separate from the training dataset?

> *Keeping the test set separate prevents the model from being evaluated on examples it has already seen during training. This gives a more honest*

estimate of how well the model can recognize unseen images.

Part 3 — Simple Neural Network

3.1 Model Architecture

Present the architecture of your neural network.

Input (64 × 64 × 3) → Flatten (12,288 features) → Dense 256 ReLU → Dropout 0.3 → Dense 128 ReLU → Dropout 0.3 → Dense 9 Softmax.

3.2 Model Configuration

Complete the table.

Component	**Configuration**
Input Image Size	64 × 64 × 3
Flatten	12,288 features
Dense Units	256 → 128 → 9
Activation Function	ReLU (hidden layers); Softmax (output)
Optimizer	Adam
Loss Function	Categorical crossentropy
Number of Epochs Batch Size	20 32

3.3 Understanding Flatten

Explain what happens when an image passes through the Flatten layer.

The Flatten layer changes the 64 × 64 × 3 image tensor into a one-dimensional vector. It does not perform feature extraction; it simply rearranges *the 12,288 pixel values into a single list so the data can be passed to the dense layers.*

Part 4 — Model Training

4.1 Training Configuration

Record your training configuration.

Parameter	Value
Epochs	20
Batch Size	32
Optimizer	Adam
Learning Rate	
Training Images	6,300
Validation Images	1,350

4.1 Training and Validation Curves

Figure 3. Training and Validation Loss

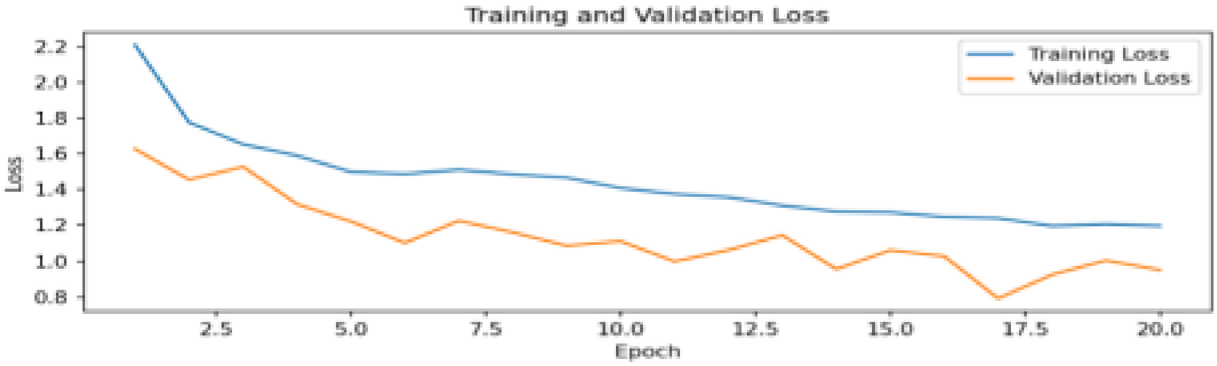
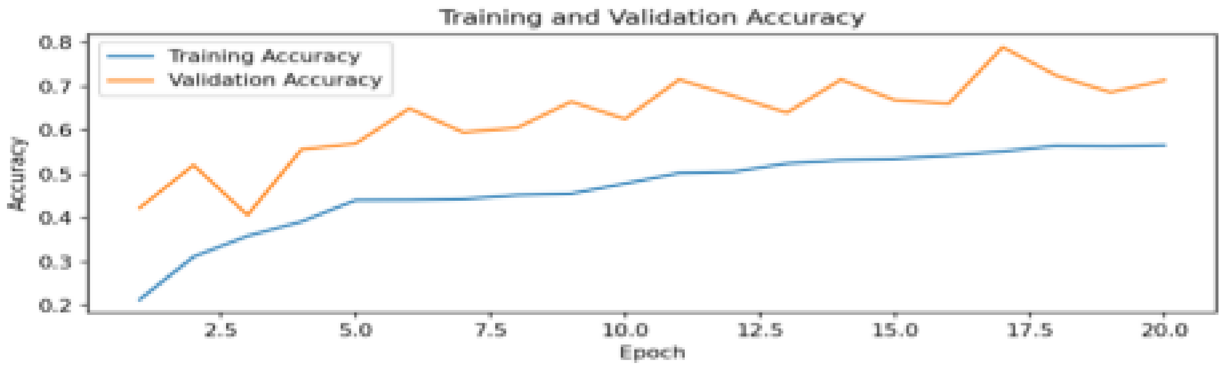


Figure 4. Training and Validation Accuracy



4.2 Interpretation

What do the training curves tell you about your model?

> *Training accuracy increased from 0.2140 in epoch 1 to 0.5659 in epoch 20, while validation accuracy fluctuated and reached 0.7896 at epoch 17 before ending at 0.7141. Training loss generally decreased, while validation loss also fluctuated. The gap and fluctuations suggest that the model learned useful patterns but did not improve smoothly on validation data throughout training.*

Part 5 — Experiment: Image Resolution

5.1 Experiment Objective

The objective of this experiment is to determine:

> **Does image resolution affect the performance of a simple neural network?**
>
> *The experiment is intended to determine whether changing image resolution affects the performance of the same simple neural network.*

The same model and dataset will be used while changing only the image resolution.

5.2 Experimental Conditions

Experiment	**Image Resolution**	**Input Features**
Experiment 1	32 × 32 × 3	3,072
Experiment 2	64 × 64 × 3	12,288
Experiment 3	128 × 128 × 3	49,152

Calculate the number of input features for each resolution.

Part 6 — Experiment Results

6.1 Performance Comparison

Complete the following table.

Resol uti on	**Input Features**	**Parame te rs**	**Training Accuracy**	**Validation Accuracy**	**Test Accur ac y**	**Traini ng Time**
32 × 32	3,072					
64 × 64	12,288	3,180,041	56.59%	71.41% (epoch 20)	71.85%	≈26.9 s*
128 × 128	49,152					

6.2 Accuracy Comparison

Create a graph comparing the test accuracy of the three experiments.

Figure 5. Test Accuracy by Image Resolution

N/A

6.3 Training Time Comparison

Create a graph comparing the training time of the three experiments.

Figure 6. Training Time by Image Resolution

N/A

6.4 Experimental Observation

Describe what happened when the image resolution increased.

The model was trained at 64 × 64 × 3, but it does not actually run the 32 × 32 and 128 × 128 experiments. Therefore, the effect of increasing image resolution cannot be concluded from the recorded experiment results. The only confirmed result is that the 64 × 64 input produced a test accuracy of 71.85% with 3,180,041 trainable parameters.

Part 7 — Model Evaluation

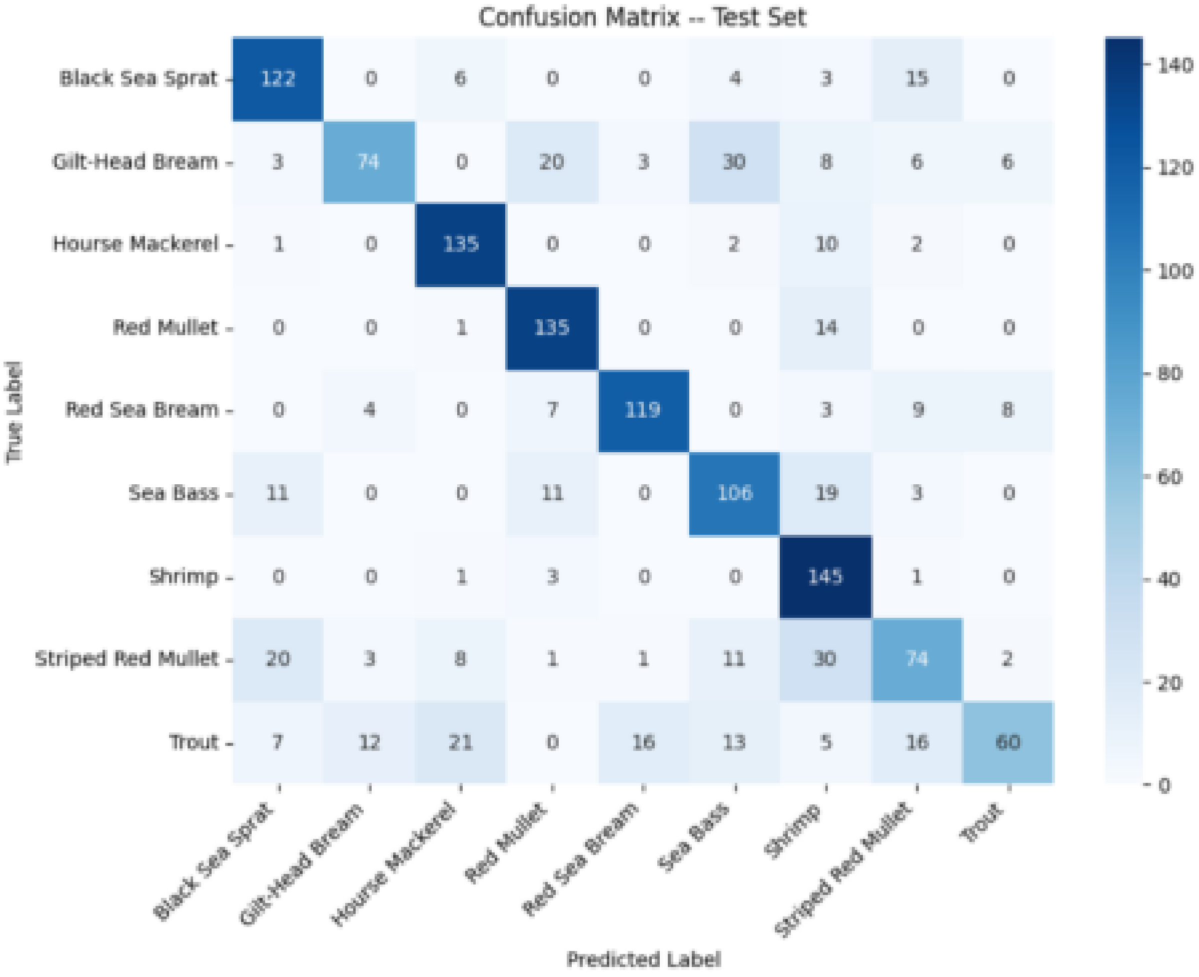
7.1 Test Performance

Report the performance of your final/best model.

Metric	Result
Test Loss	0.9511
Test Accuracy	0.7185 (71.85%)

7.2 Confusion Matrix

Figure 7. Confusion Matrix



7.3 Classification Performance

Insert your classification report.

Class	Precision	Recall	F1-Score	Support
Black Sea Sprat	0.74	0.81	0.78	150
Gilt-Head Bream	0.80	0.49	0.61	150
Hourse Mackerel	0.78	0.90	0.84	150
Red Mullet	0.76	0.90	0.83	150
Red Sea Bream	0.86	0.79	0.82	150

Sea Bass	0.64	0.71	0.67	150
Shrimp	0.61	0.97	0.75	150
Striped Red Mullet	0.59	0.49	0.54	150
Trout	0.79	0.40	0.53	150

7.4 Class Performance

Best Performing Class

Class: Hourse Mackerel

Reason/Observation:

F1-score = 0.84, the highest class F1-score in the report. Its recall is also high at 0.90, indicating that most Hourse Mackerel test images were correctly identified.

Most Difficult Class

Class: Trout

Reason/Observation:

Trout has the lowest recall at 0.40 and an F1-score of 0.53. The confusion matrix shows notable confusion with Hourse Mackerel (21), Red Sea Bream (16), and Striped Red Mullet (16).

Part 8 — Prediction Analysis

8.1 Correct Predictions

Display examples of correctly classified images.

Figure 8. Correct Predictions

N/A

Briefly discuss what these images have in common.

8.2 Incorrect Predictions

Display examples of incorrectly classified images.

Figure 9. Incorrect Predictions

N/A

8.3 Prediction Errors

Choose at least **three incorrect predictions** and analyze them.

Image	Actual Class	Predicted Class	Possible Reason
1	Gilt-Head Bream	Sea Bass	The confusion matrix shows 30 Gilt-Head Bream images predicted as Sea Bass, suggesting visual similarities that the flattened-pixel model could not distinguish reliably.
2	Striped Red Mullet	Shrimp	30 Striped Red Mullet images were predicted as Shrimp, which may be due to similarities in their shape or visual features that the simple model could not distinguish.

3	Trout	Hourse Mackerel	21 Trout images were predicted as Hourse Mackerel, suggesting that these two classes have visual characteristics that are difficult for the simple neural network to differentiate.

Discussion

Why do you think the neural network confused these classes?

The greatest errors are between classes of fish that are visually similar. The network does not have any special design elements to learn local shapes, edges, textures or spatial patterns due to flattening raw pixels and consecutive dense layers. This is a likely reason why class confusion occurs when classes of similar appearance are given. The reasons given above are the interpretation of the confusion matrix and are not the actual reasons for the misclassified images.

Part 9 — Reflection

9.1 What Did You observe?

The experiment revealed that a simple fully connected neural network can be trained to extract meaningful patterns from fish images, but the performance is not as good as would be expected by a more image-oriented architecture. The model achieved the training accuracy of 56.59% and the test accuracy of 71.85% after 20 epochs. There were a few classes identified very well and many others that were often misidentified. The training curves were also erratic, indicating that the performance during validation was not always increasing.

Part 10 — Conclusion

Write a concise conclusion summarizing:

1. The dataset you used.
2. The performance of your simple neural network.
3. The effect of image resolution.
4. The major problems or limitations encountered.
5. Your overall answer to the question:

Can a Simple Neural Network Recognize Images?

Conclusion:

A dataset of 9 classes with 9,000 images of fish, and trained a simple neural network with 64 × 64 × 3 RGB inputs. The model consisted of a Flatten layer, two dense layers of 256 and 128 ReLU units, and a Softmax output layer of 9 units. The model was trained for 20 epochs with a batch size of 32, and the test loss and test accuracy were 0.9511 and 71.85%, respectively. The results demonstrate that a simple neural network can be used to successfully identify fish images to a certain extent, but the success rate differs significantly from one class to another. It is not possible to fully evaluate the planned resolution experiment as it is only possible to access the 64 × 64 run; runs of 32 × 32 and 128 × 128 have not been accessed. Other features are not included, such as using a Flatten layer rather than convolutional feature extraction and absence of saved single correct/incorrect prediction examples.

Overall answer: *Yes. A simple neural network can recognize images, as demonstrated by the 71.85% test accuracy, but the experiment also shows that a simple architecture has clear limitations for image classification.*

Part 11 — Appendix

Include the following:

Appendix A — Google Colab Notebook

Provide the link to your completed Google Colab notebook.
Colab Link:
https://colab.research.google.com/drive/1pPkGMyKcpRw83ZhLtXMvgaOruhMSc8Tu?usp=sharing

Appendix B — Additional Results

Include any additional graphs, tables, screenshots, or results that support your discussion.

N/A